

The next 10 years in technology

I've always been interested in technology and am now a full professor for robotics at the University of Manitoba in Winnipeg, Canada. I've been involved in robotics research, competitions and education for more than 15 years now.

I am convinced that the next 10 years will bring the age of personal robotics. Note that I don't mean automation in manufacturing — expensive special purpose machines performing repetitive tasks to build cars and various other products. I'm talking about affordable, flexible, adaptive and intelligent robots that people use in many new and exciting ways that we've not even begun to explore yet.

The robotic age will have an even larger impact on our society than the personal computer (PC) in 1980s, the internet in the 1990s and mobile computing and social media in the 2000s. The PC is only 30 years old, yet there is hardly anyone in the world that has not been affected by its introduction. The PC has fundamentally changed our society — the way we work, play and interact with each other.

So far, though, PCs have been limited to a virtual world. They can fetch data from a database, push numbers around in a spreadsheet or send messages, but can't yet manipulate the physical world at large.

But with personal robotics, these computers can now reach out and touch our physical world. The resulting changes to our society will make the PC revolution look small compared to the tsunami of change that will come with personal robotics.

A glimpse of these exciting new possibilities can be seen at the intelligent robot competitions that have gained great popularity worldwide. For example, the RoboCup competition has the goal of beating the human world champion by the year 2050 [1]. FIRA HuroCup is the toughest challenge for humanoid robots where a

single robot has to sprint, run a marathon, traverse an obstacle course, lift weights, walk over uneven terrain, play basketball, kick soccer penalties and climb walls [2]. It's refreshing to see that not just top universities from around the world participate in these competitions, but also hobbyists, which shows that average persons are getting involved. Recently, companies such as iRobot have produced successful commercial robots that can vacuum a floor in your house.

From these humble beginnings, I believe that in 10 years you'll have personal robots that monitor your children at home, substitute for teachers in classrooms, act as your avatar while attending

a meeting or family gathering half way around the world, drive you to work in your car, wash and fold your clothes, clean up your messy house and rescue people from a burning building.

Many of the tasks above can already be done by special purpose robots, but those robots are too expensive and require a team of experts to run. There's a lot of work to be done to make these robots personal, i.e. reducing their cost and making them robust enough for peoples' homes and flexible enough to perform many tasks.

The increased use of computers and robotics also means that programming literacy or the ability to program your ever more powerful and complex personal robots will become a crucial element in our education similar to reading, writing and arithmetic.

Some say that those predictions are unrealistic and that it will take much longer, but I'm reminded of the Wright brothers who flew basically a powered kite for a little over 30m in 1903, but less than 50 years later, in 1947 Chuck Yeager broke the sound barrier in a jet plane. Technology is advancing at an incredible rate.

Like any technology, not all uses of robotics will have a positive impact on our society. If you build robotic firemen and rescue workers, then others will turn them into robotic soldiers to kill. Use and abuse of technology, however, is in my view a societal problem and needs to be discussed in this context.

I, therefore, am trying to encourage my and other students of engineering or science to get involved in politics to make sure that their hard work is not being abused. I also believe that sound scientific knowledge is crucial for understanding the most pressing problems of our times, such as sustainable development and climate change. 



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REFERENCES:

<http://www.robocup.org>; <http://www.fira.tw>